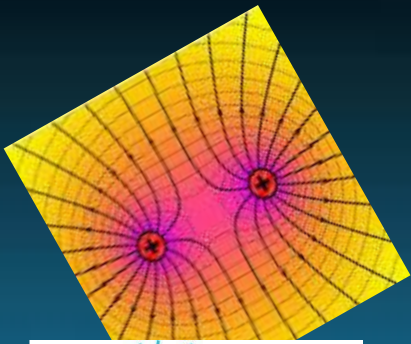
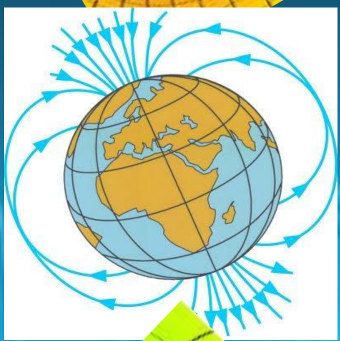
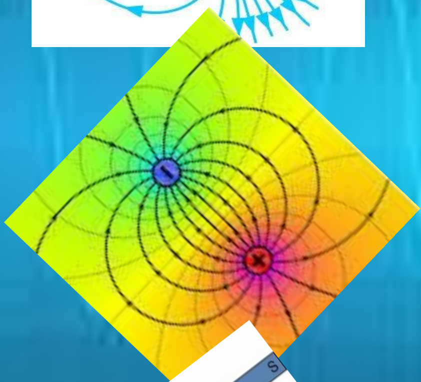
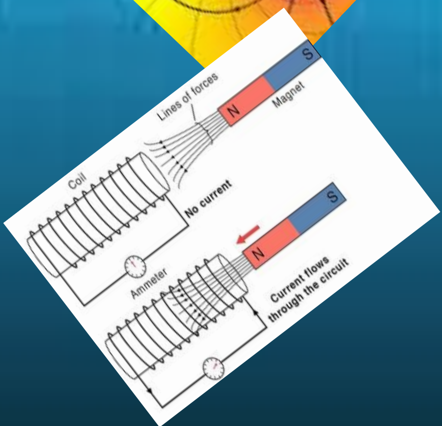
The Federal Polytechnic Mubi, Nigeria
Department of Electrical and Electronic Engineering Technology









EEC 431

Electromagnetic Field Theory

Higher National Diploma (HND)

Lecture notes prepared by

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1 INTRODUCTION

1.1 Electromagnetic Field Theory

Electromagnetic field theory is based on a few fundamental principles of electric & and magnetic fields and waves that are exploited and used to fuel some of the most innovative technological breakthroughs in the history of mankind. Many necessary technologies which include the use of electricity for generation, transport and conversion of energy; communication systems such as satellites, mobile, radio and televisions; insulation coordination in power systems; Radar Technologies; Instrumentations and a number of electrical and electronic devices and systems.

Electromagnetic fields are a combination of invisible electric and magnetic fields of force. They are generated by natural phenomena such as the earth's magnetic field or thunderclouds, etc. They can also be generated by human activities, mainly through the use of electricity.

1.2 Main Fundamental Concepts of Electromagnetic Fields

1.2.1 Electric Charge

Every constituent of matter comprises atoms containing electric charges with a value that can be positive, negative, or zero. For example, electrons are negatively charged, and atomic nuclei are positively charged.

1.2.2 Electric Fields

When an electric charge is present, it creates an electric field around it which gives an idea about the space or region around the charge. This field can exert a force on other charges in the vicinity.

Electric field is categorized into two – static electric fields and time varying electric fields. Static electric fields, also known as electrostatics are created by charges that are fixed in space. Figure 1 illustrates some static fields and their patterns. These patterns are just imaginary lines used to give idea about the region of energy around the electric charge(s). The best-known and most powerful display of static electric fields occurs in nature as lightning and static shocks such as those created due to friction with clothes and other materials. Time-varying electric field is obtained when the current continuously changes its direction as is the case with the technically generated 50 Hz alternating current.

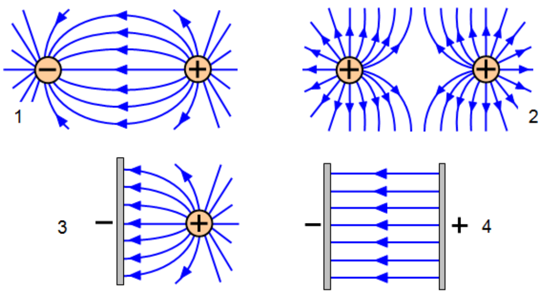
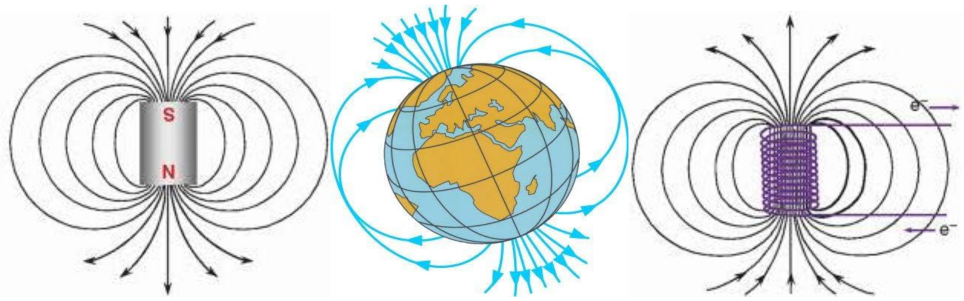


Figure 1 Static Electric Fields and Patterns

1.2.3 Magnetic Field

Apart from magnetic materials such as magnetite, moving electric charges also create magnetic fields. These fields can interact with other magnetic or electric charges to produce motion or electricity.

Just like an electric field, the magnetic field can also be of a static (DC field) type or time-varying type. Static magnetic fields, also known as magnetostatic fields are created by magnets or direct current (DC) as shown in Figure 2 below. Dynamic magnetic fields are obtained by alternating current (AC) at a certain frequency. The mutual effect of both dynamic magnetic and electric fields is shown in Figure 3.



(a) Bar magnet (b) Geomagnetic field (c) Coiled conductor carrying DC

Figure 2 Static Magnetic Fields

1.2.4 Time-varying fields (or electromagnetic fields)

The field produced due to accelerated charges or time-varying current (AC) is called a time-varying field as mentioned above.

The Static fields are not dependent on each other. However, the time-varying electric and magnetic fields are dependent on each other as shown in figure 3. In other words, a time-varying electric field is produced by a time-varying magnetic field and a time-varying magnetic field is produced by a time-varying electric field.
The first concept was experimentally introduced by Michael Faraday and the second was theoretically introduced by James Clerk Maxwell.

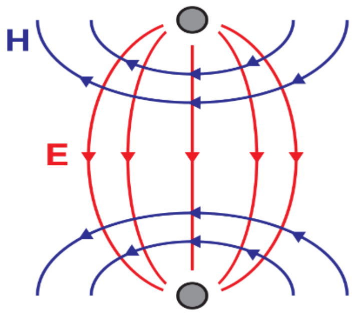


Figure 3 Time-varying (electromagnetic) fields

1.2.5 Electromagnetic Wave

When electric and magnetic fields oscillate at right angles to each other, they produce electromagnetic waves. These waves can travel through space and carry energy from one place to another.

1.3 The Objective of the Course Content

The major objective of the course curriculum is to equip a student engineer with a scientific background concerning electromagnetic field theories which are the fundamental framework for many branches of electrical engineering and some other related disciplines.

1.4 The Scope of the Course

In some detail, the course will cover relevant theories and laws which are important to the understanding of the following aspects:

- Principles and applications of static electric fields (electrostatics)
- Principles and applications of static magnetic fields (magnetostatics)
- Principles and applications of time-varying electromagnetic fields
- Principle and applications of plane waves

2 STATIC ELECTRIC FIELDS (ELECTROSTATICS)

2.1 Coulomb's Law

Coulomb's law is a mathematical description of the electric force between two point charges. The word point charge means the size of linear charged bodies is very small against the distance between them. Therefore, they are considered point charges as it becomes easier to calculate the force of attraction/ repulsion between them. The statements of the law are as follows:

- like charges repel each other and unlike charges attract each other (the attraction or repulsion ***acts along the line*** between the two charges).
- The electrostatic force between two-point charges q_1 and q_2, separated by a distance R, is directly proportional to the product of the charges and inversely proportional to the square of the distance between them.

$$F = k\frac{q_1 q_2}{R^2} \tag{1}$$

Where:

- ✓ The force, F expressed in newton (N).
- ✓ The charges q_1 and q_1 in Coulombs (C).
- ✓ The separation, R in meters (m) and
- ✓ k is a constant $= \frac{1}{4\pi\varepsilon}$; ε is the permittivity of the dielectric medium $= \varepsilon_o\varepsilon_r$ (ε_o is the permittivity of free space given by $8.85 \times 10^{-12}\ F/m$ and ε_r, permittivity of a given medium).

2.1.1 Coulomb's law in a vector form

from the first statement of charge interaction (opposites attract and likes repel), it can be deduced that Electrical Force is a vector quantity as it has both magnitude and direction. The Coulomb's law in equation 2.1 can, therefore, be rewritten in the form of vectors. In this case, the vector "F" is denoted by $\vec{\boldsymbol{F}}$ and the vector R by $\vec{\boldsymbol{R}}$ and so on.

In the diagram of Figure 4 below, objects A and B have like charges causing them to repel each other. Thus, the force on object A is directed leftward (away from B) and the force on object B is directed rightward (away from A). On the other hand, objects C and D have opposite charges causing them to attract each other. Thus, the force on object C is directed rightward (toward object D) and the force on object D is directed leftward (toward object C).

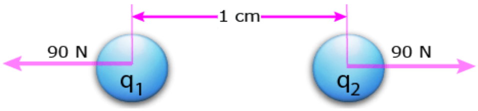
Figure 4 Determining the direction of the electrical force vector

Solved example 1

The force between two identical charges separated by 1 cm is equal to 90 N. What is the magnitude of the two charges? (assume free space).

Solution:

First, draw a force (or vector) diagram of the problem illustrating the situation as shown below.



From the Coulomb's law equation:

$$F = K\frac{q_1 q_2}{R^2} = K\frac{q^2}{R^2} \text{ (since the charges are identical)}$$

$$\therefore \quad q^2 = F\frac{R^2}{K}$$

$$q = \sqrt{F\frac{R^2}{K}}$$

$$q = \sqrt{\frac{(90N)(0.01m)^2}{9\times 10^9\ N.m^2/C^2}} = \pm 1.0\times 10^{-6} C$$

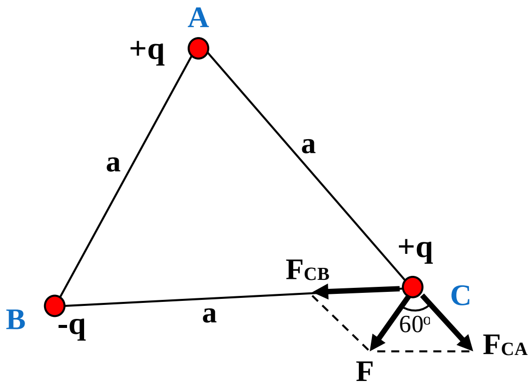
This equation has two possible answers. The charges can both be positive or negative and the answer will be the same for the repulsive Columb force over a distance of 1 cm.

Solved example 2

Three charges of magnitude 100 µC each are located in a vacuum at the corners A, B and C of an equilateral triangle measuring 4 m on each side. If the charges at A and C are *positive* and the charge at B is *negative*, what is the magnitude and direction of the total force on the charge at C?

Solution:

The situation is shown in the figure below.



Let us consider the forces acting on C due to A and B.

Now, from Coulomb's law, the force of repulsion on C due to A, i.e., F_{CA} in the direction of AC, is given by:

$$F_{CA} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q\times q}{a^2} \text{ along CA.}$$

The force of attraction on C due to B, i.e., F_{CB} in direction CB, is given by:

$$F_{CB} = \frac{1}{4\pi\varepsilon_0}\cdot\frac{q\times q}{a^2} \text{ along CB.}$$

Thus, the two forces are equal in magnitude. The angle between them is 120°. The resultant force F is given by:

$$F = \sqrt{F_{CA}^2 + F_{CB}^2 + 2F_{CA}\times F_{CB}\cos120^{o}}$$

$$= \frac{q^2}{4\pi\varepsilon a^2}$$

$$= \frac{9\times10^9(100\times10^{-6})^2}{4^2} = \mathbf{5.625\ N}$$

Exercise

Determine the magnitude of a test charge q, 12 cm from a point charge of magnitude -9 μC if it experiences an electrostatic force 22.275 N. is this force repulsive or attractive.

2.2 Electric field intensity (or Field strength)

The coulomb's law gives the mathematical relationship between electrical forces, charges and their separation in an electric field. However, to describe how this force is felt by an individual charge placed at certain location within the field, the term field intensity is used.

The **electric field intensity E,** at a point is the force experienced by a unit positive charge placed at that point.

Mathematically, $$\vec{E} = \frac{\vec{F}}{q} \text{ (in N/C or V/m)} \qquad (2)$$

- The electric field intensity due to a positive charge is always directed away from the charge, and the intensity due to a negative charge is always directed towards the charge.

- The intensity of the electric field due to a test charge q placed at a point d units away from a source charge Q, is given by the expression:

$$\vec{E} = k\frac{Q}{\vec{d}^2} \text{ (N/C)} \qquad (3)$$

Note that equation 3 above shows that the field strength is dependent only upon the quantity of charge on the source charge, Q and the distance of separation d, from the source charge.

- The intensity of the electric field at any point due to several charges is equal to the vector sum of the intensities produced by the separate charges.

Solved examples

(1) In the vicinity of point charge Q, we place a 0.2 μC-charge so that a force of $5 \times 10^{-5}\ N$ is applied to it due to the charge Q. Find the electric field produced by this unknown charge Q.

Solution: The Coulomb force exerted on a test point charge q at any point is related to the electric field (due to an unknown charge Q) at that point by:

$$\vec{F} = q\vec{E}$$

Therefore, the magnitude of the electric field is obtained as

$$E = \frac{F}{q}$$

$$= \frac{5 \times 10^{-5}}{0.2 \times 10^{-6}} = 250\ N/C$$

The electric field problems are closely related to Coulomb's force problems.

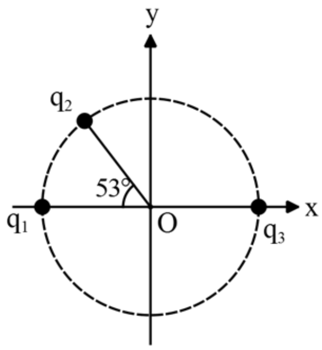
(2) What is the magnitude and direction of the electric field due to a point charge of 20 μC at a distance of 1 meter away from it? ($k = 9 \times 10^9\ \frac{N \cdot m^2}{C^2}$).

Solution: The magnitude of the electric field due to a point charge q at a distance r from it is given by $E = k\frac{q}{r^2}$

Substituting the numerical values into this, we obtain: $E = \frac{(9 \times 10^9)(20 \times 10^{-6})}{1^2} = 1.8 \times 10^5\ N/C.$

The sign of the charge determines the direction of the electric field. In this case, since the charge is positive, the electric field at every point is away from it.

(3) In the figure below, three equal charges $q_1 = q_2 = q_3 = +4\mu C$ are located on the perimeter of a sphere of diameter 12 cm. Find the net electric field, in terms of unit vectors $\hat{\imath}, \hat{\jmath}$ at the centre of the sphere.



Solution:

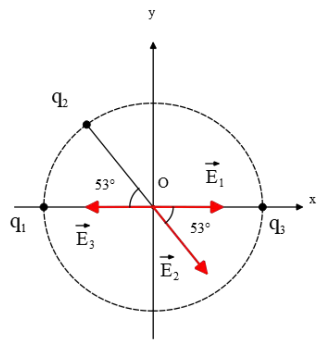
Reasoning: The electric fields of charges q_1 and q_3 at the centre of the sphere are equal in magnitude and opposite in direction. Since the magnitude of charges is the same $q_1 = q_3$ and they are located at an equal distance from the centre, we can use the definition of the electric field to determine that:

$$E_1 = k\frac{q_1}{r^2} = E_3, \qquad \vec{E}_1 = -\vec{E}_3$$

Therefore, the resultant of electric field vectors at point O is equal to the field $\vec{E}_2$, using the superposition principle of fields: $\vec{E}_{net,O} = \vec{E}_1 + \vec{E}_3 + \vec{E}_2 = \vec{E}_2$ (since $\vec{E}_1 + \vec{E}_3 = 0$)

However, the electric field $\vec{E}_2$ lies in the fourth quadrant along the radius of the sphere, making an angle of 53^o relative to the $+x$ axis as shown in the figure below. In such cases we must decompose the vector into its

components in *x* and *y* directions.



First, use the definition of the electric field of a point charge to find the magnitude of $\vec{E}_2$ as $10^7\ N/C$.

Next, we can decompose the vector $\vec{E}_2$ using elementary geometry as follows:

$$\vec{E}_2 = |\vec{E}_2| cos53^o(\hat{\imath}) + |\vec{E}_2| sin53^o(-\hat{\jmath})$$

$$= \quad 10^7(0.6)(\hat{\imath}) + 10^7(0.8)(-\hat{\jmath})$$

$$= \quad 6\times10^6\hat{\imath} - 8\times10^6\hat{\jmath} \quad \textbf{(N/C)}$$

2.2.1 Electric field lines

A useful means of **visually** representing electric field (and of course electric field strength/density) is through the use of electric field lines of force. Since electric field is a vector quantity, it can be represented by a vector arrow as illustrated in figure 1 (chapter 1). Electric field lines reveal information about the direction (and the strength) of an electric field within a region of space. The following are the properties are the properties of and rules of drawing electric filed lines:

- The field lines begin either at +ve charges or infinity and end either at -ve charges or at infinity: object (a).
- The start point of the field line is at the positive charge and end at the negative charge: object (a)
- The number of filed lines depends on the charge (higher the charge, higher the field lines): object (b).
- When the field is stronger, the filed lines are closer to each other and vice versa: object (c).
- The field lines never intersect at each other: object (d).
- The field lines are perpendicular to the surface of the charge.

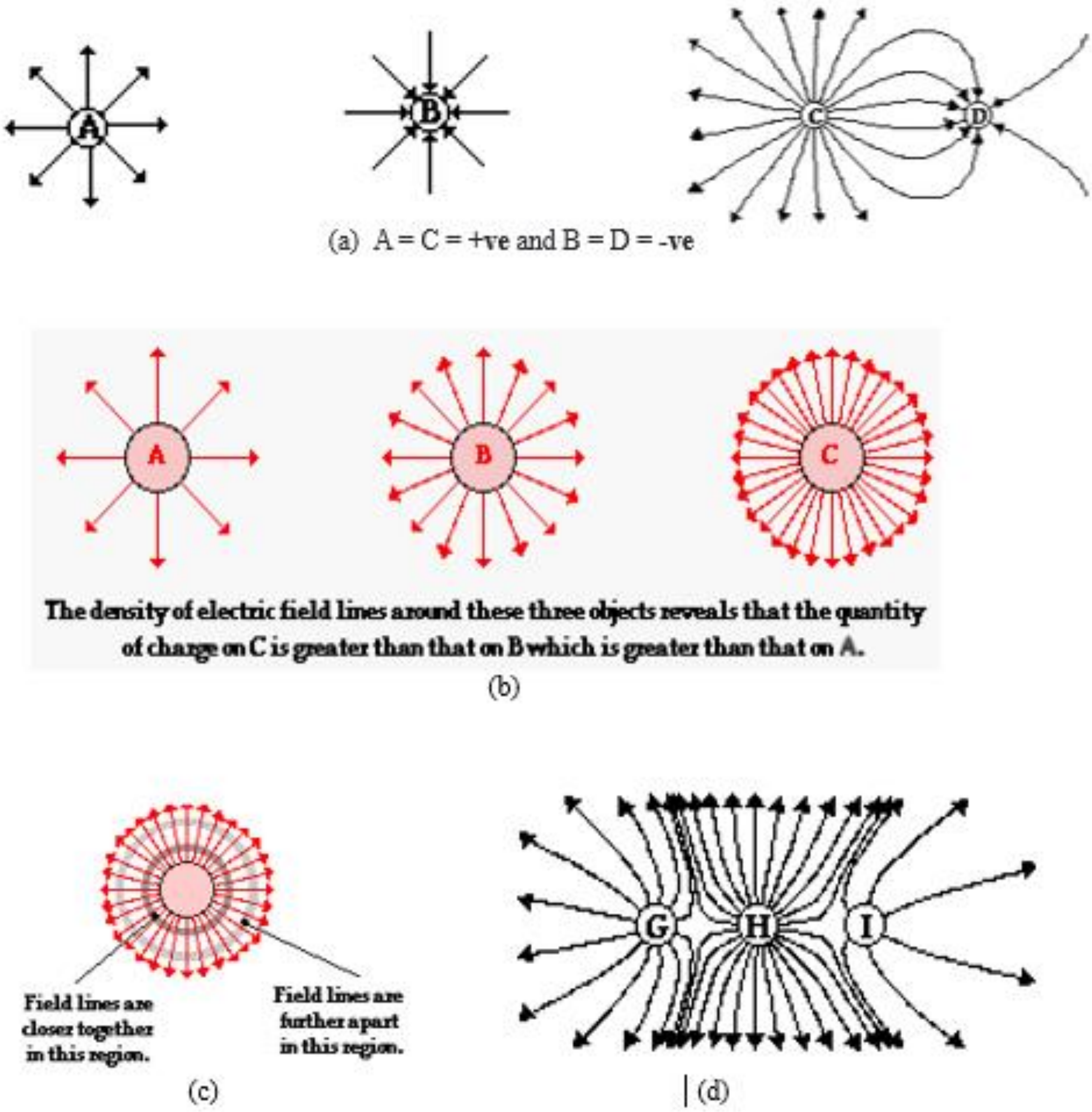


Figure 5 **Electric field line vectors for locations around +ve and -ve objects**

2.3 Electric flux

Electric flux Φ_E or just Φ, is defined as the total number of electric lines of force emanating from a charged body. Through a surface area A, the electric flux (Φ) is the dot product of the electric field vector $\vec{\mathbf{E}}$, and the area vector $\vec{\mathbf{A}}$, as shown in figure 6.

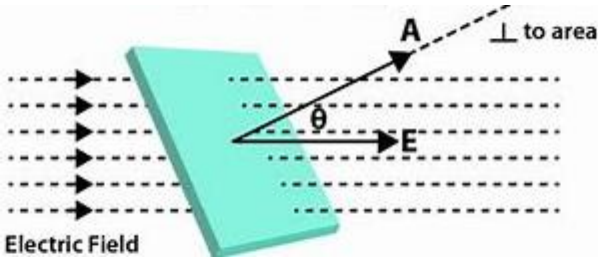


Figure 6 Electric flux through a surface area vector A.

Therefore, electric flux $\Phi, = \vec{\mathbf{E}}.\vec{\mathbf{A}} = \mathbf{EA}\cos\theta$. Where θ is the angle between the electric field lines and the normal (perpendicular) to the surface area. In an integral form,

$$\Phi = \int \mathbf{E}d\mathbf{A}\cos\theta \ (\frac{Nm^2}{C}) \tag{4}$$

2.3.1 Electric flux density

As the number of lines of force emanated from a positive charged body is numerically equal to the charge of the body measured in coulomb, the unit of flux is also taken as coulomb. If Q is charge of a body and Φ is the flux from the body then we can write:

$$Q = \Phi \text{ (Coulombs)} \tag{5}$$

Electric flux density at any point in the electric field of a body is the number of lines of force passing (perpendicularly) through a unit surface area at that point. Recall equation 3, the expression of electric field at a point given by:

$$E = k\frac{Q}{d^2} = \frac{Q}{4\pi\varepsilon d^2}$$

Since Q = Φ, we have $E = \frac{\Phi}{4\pi\varepsilon d^2}$. This can be rewritten as:

$$\varepsilon E = \frac{\Phi}{4\pi d^2} \tag{6}$$

This is the expression of flux per unit area since $4\pi d^2$ is the surface area of the imaginary sphere of radius d which is the distance from the centre of the charge. This is called **electric flux density** at the said point. We generally denote it with an English letter *D*. Therefore,

$$D = \varepsilon E \tag{7}$$

2.4 Gauss's law

Gauss's law establishes a relationship between the enclosed charge Q and the electric flux linked to the enclosing surface.

The law states that the electric flux (Φ_E) out of an *imaginary closed surface* is equal to the charge (Q) enclosed by the surface divided by the permittivity of free space (ε_o). *An imaginary closed surface is referred Gaussian surface.*

From the definition of flux in equation 4, Gauss's law can mathematically be expressed as:

$$\oint \mathbf{E}\mathrm{d}\mathbf{A}\cos\theta = \frac{\mathrm{Q}}{\varepsilon_o} \tag{8}$$

2.4.1 Applications of Gauss's Law

Gauss's law is applied mainly to determine the electric field **E**, due to finite symmetric charge distribution in cases such as:

i. Uniformly charged **infinite straight wire.**
ii. Uniformly charged **infinite plane sheet.**
iii. Uniformly charged **thin** conducting or non-conducting **spherical shell.**

However, before we consider these cases, let us take the simplest and the most basic case (i.e., a point charge, *q*) to determine **E** using gauss's law as shown in Figure 7**.**

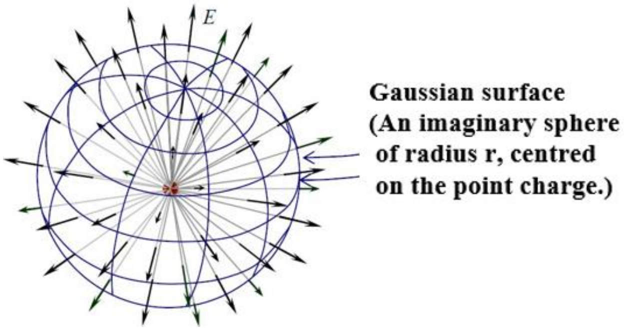


Figure 7 A point charge with spherical symmetry

We consider any point located on the imaginary Gaussian surface of radius r from the point charge as shown in the figure. Applying Gauss's law, we have:

$$\Phi_E = \oint \mathbf{E}\mathrm{d}\mathbf{A}\cos\theta = \frac{\mathrm{q}}{\varepsilon_o}$$

The electric field intensity being *normal* and constant at every point on the surface, we have:

$$\mathrm{E}4\pi r^2 = \frac{\mathrm{q}}{\varepsilon_o} \text{ or}$$

$$\mathrm{E} = \frac{\mathrm{q}}{4\pi\varepsilon_o r^2} = \frac{k.\mathrm{q}}{r^2}$$

This produces the same equation obtained from Coulomb's law.

A. Electric field due to uniformly charged infinite straight wire

We consider any point located at a distance r from the wire. If have several of such a point with equal distance, we can imagine a cylindrical Gaussian surface with a height *h* and a radius *r* around the wire. The cylinder is closed from both left and right and is parallel to the wire at every point as shown in figure 8.

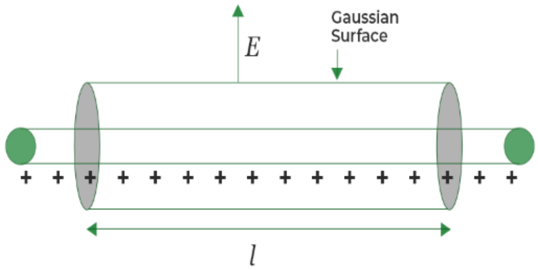


Figure 8 Electric field due to uniformly charged infinite straight wire

We know that for this uniformly distributed positive charge of the wire, the electric field will be radially outward, perpendicular to the curve surface of the cylinder. Therefore, using Gauss's law where,

$$\Phi_E = \oint \mathbf{E}d\mathbf{A}\cos\theta = \frac{\mathrm{q}}{\varepsilon_o}$$

The total electric flux linked to the cylindrical Gaussian surfaces is, therefore, given by:

$$\Phi_{E\,(total)} = \Phi_{E\,(left\ circular\ surface)} + \Phi_{E\,right\ circular\ surface)} + \Phi_{E\,(curved\ circular)} = \frac{\mathrm{q}}{\varepsilon_o}$$

$$\oint \mathbf{E}d\mathbf{A}\cos 90^o + \oint \mathbf{E}d\mathbf{A}\cos 90^o + \oint \mathbf{E}d\mathbf{A}\cos 0^o = \frac{\mathrm{q}}{\varepsilon_o}$$

$$\oint \mathbf{E}d\mathbf{A} = \frac{\mathrm{q}}{\varepsilon_o}$$

$$\mathbf{E} \cdot 2\pi\mathrm{rh} = \frac{\mathrm{q}}{\varepsilon_o}$$

$$\mathbf{E} = \frac{\mathrm{q}}{(2\pi\mathrm{rh})\varepsilon_o}$$

But we know that the linear charge density $\lambda, = \frac{charge\ (q)}{unit\ lenght\ (l)} = \frac{q}{h}$ or $q = \lambda h$. Therefore, the field intensity can be more appropriately written as:

$$\mathbf{E} = \frac{\lambda}{2\pi\varepsilon_o \mathrm{r}} \qquad (9)$$

It can be observed that there is an inverse relationship between the field intensity E, and the distance r from the wire. As we move away from the wire, the intensity decreases and vice versa.

B. Electric field due to uniformly charged infinite plane sheet

We will calculate the electric field due to infinitely large plane sheet with a uniformly distributed charges on its surface. As shown in figure 9, We can take a small portion of the sheet and extend equally on both sides a distance of r away from the it to form a cylindrical Gaussian surface. As the electric field pokes out perpendicularly to the sheet, it becomes parallel to the Gaussian surface as shown.

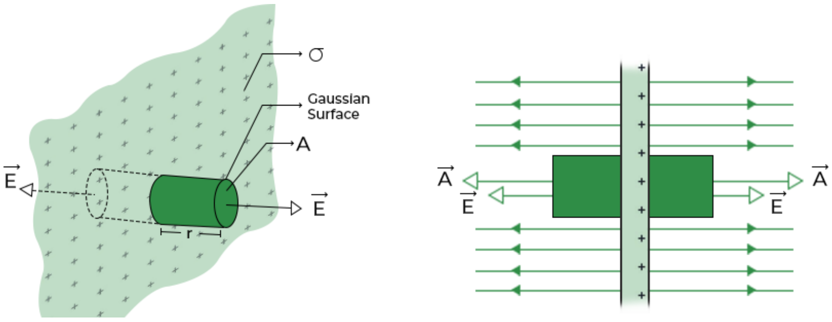


Figure 9 Electric field due to infinite plane sheet

The electric flux linked to the entire Gaussian surface is similar to the first case and is given by:

$$\Phi_{E\,(total)} = \Phi_{E\,(left\ circular\ surface)} + \Phi_{E\,(right\ circular\ surface)} + \Phi_{E\,(curved\ circular)} = \frac{q}{\varepsilon_o}$$

$$\oint \mathbf{E}\mathrm{d}\mathbf{A}\cos 0^o + \oint \mathbf{E}\mathrm{d}\mathbf{A}\cos 0^o + \oint \mathbf{E}\mathrm{d}\mathbf{A}\cos 90^o = \frac{q}{\varepsilon_o}$$

$$\oint \mathbf{E}\mathrm{d}\mathbf{A} + \oint \mathbf{E}\mathrm{d}\mathbf{A} = \frac{q}{\varepsilon_o}$$

$$2\boldsymbol{E}\pi R^2 = \frac{q}{\varepsilon_o}$$

$$\therefore \quad \boldsymbol{E} = \frac{q}{(2\pi R^2)\varepsilon_o}$$

R is the radius of the circular surface which is the same for both surfaces. Now the charge q, responsible for the flux linked to the cylindrical Gaussian surface is a circular path through which the imaginary cylinder cuts the sheet as illustrated in the figure. As such, only the charge within circular surface will interact with the Gaussian surface. In the case of a surface, we use the term surface charge density, σ to simplify the relation as follows.

The surface charge density, $\sigma = \frac{charge\ (q)}{unit\ circular\ area} = \frac{q}{\pi R^2}$ (giving $q = \pi R^2 \sigma$)

$$\therefore \quad \boldsymbol{E} = \frac{\sigma}{2\varepsilon_o} \qquad (10)$$

It can be noted from equation 10 that the electric field intensity does not depend on distance from the sheet.

With Gauss's law, similar relationship can easily be obtained for electric fields such as Uniformly charged thin conducting or non-conducting spherical shell, etc.

2.5 Energy Stored in an Electric Field